

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

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Abstract

001 Evaluating AI/ML systems responsibly requires prac-
002 tices for testing, stress-testing, auditing, comparing,
003 and interpreting specific capabilities—not only aggre-
004 gate benchmark scores. This paper proposes such a
005 practice for spatial reasoning in vision–language mod-
006 els (VLMs): a diagnostic audit of object-relative orien-
007 tation, a capability that aggregate accuracy can hide.
008 On Visual Spatial Reasoning (VSR), moving from a
009 compact 2.2B model to a larger 7B VLM substantially
010 improves overall, depth, horizontal, and containment
011 accuracy, but orientation changes only modestly. Ex-
012 cept for structured prompting, which degrades perfor-
013 mance, trained conditions on both models remain in
014 a narrow 62–66% orientation band. We test struc-
015 tured prompting, general and targeted LoRA, hard neg-
016 atives, visual/projector adaptation, frozen and object-
017 grounded representation probes, and object-centric de-
018 composition; none of the tested configurations reliably
019 closes the gap, and a clean-label sensitivity analysis
020 shows that the pattern persists after excluding exam-
021 ples flagged as ambiguous (the single-annotator audit’s
022 reliability is assessed with a second blind annotator in
023 the supplementary).

024 Accuracy also fails to capture relational coherence.
025 LM-side adaptation significantly improves consistency
026 over zero-shot, while hard-negative training raises
027 facing/facing-away consistency from 66.0% to 77.7%
028 with facing accuracy unchanged at 68.9%; the addi-
029 tional pooled consistency gain over LM-only is not sig-
030 nificant. Cross-dataset evaluation on SITE shows that
031 VSR-trained adaptation does not transfer cleanly: it is
032 neutral overall and significantly harms official spatial-
033 relationship reasoning (75.1%→71.6%, $p = 0.004$). A
034 pre-specified orientation-vocabulary slice, frozen before
035 SITE evaluation, is analyzed as exploratory evidence
036 rather than as direct replication of the VSR construct.
037 The results point to a persistent orientation bottleneck

in which orientation is not robustly decodable by the
tested frozen-feature readouts and relational decisions
are not always propagated coherently, while also show-
ing that task-specific spatial adaptation can transfer
poorly. Each component of the audit—the diagnos-
tic ladder, the label-sensitivity check, the accuracy–
coherence separation, and the cross-dataset transfer
protocol—is specified operationally so that it can be ap-
plied by others to other capabilities, benchmarks, and
model families.

1. Introduction

This paper’s contribution is an evaluation practice as
much as a set of findings. We propose a protocol
for testing, stress-testing, auditing, comparing, and in-
terpreting a specific spatial capability across vision–
language models (VLMs), and we demonstrate it on
object-relative orientation, a relation family that ag-
gregate accuracy can hide. Spatial reasoning is central
to multimodal systems that must interact with physical
environments. A useful VLM should not only recognize
objects, but determine whether one object is in front
of another, whether it is facing another, whether two
objects are parallel, and whether equivalent or com-
plementary formulations yield logically compatible an-
swers. These abilities matter for robotics, embodied
agents, navigation, and scene understanding.

Existing work already shows that spatial relations
are difficult for vision–language systems. VSR was
designed as a controlled natural-image probe and
reported particularly poor performance on object-
orientation relations in earlier VLMs [7]. More recent
controlled benchmarks likewise find that strong VLMs
can fail on basic left/right and other grounded rela-
tions [6, 9], while broader evaluations continue to iden-
tify spatial orientation as a difficult factor [14]. Our
question is therefore not whether spatial reasoning can
fail, but whether the orientation weakness persists in

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075	modern generative VLMs of different sizes and after	128
076	targeted intervention, and what the failure signature	129
077	reveals about representation, reasoning, and transfer.	130
078	In our VSR experiments, the distinction between	131
079	relation families is striking. Moving from SmolVLM2-	132
080	2.2B-Instruct to the larger Qwen2-VL-7B-Instruct in-	133
081	creases zero-shot overall accuracy from 74.0% to 80.9%.	134
082	Horizontal reasoning rises from 70.2% to 84.8%, depth	135
083	from 68.9% to 75.2%, and containment from 83.4%	136
084	to 89.3%. Orientation, by contrast, moves only from	137
085	62.8% to 63.5%. We then evaluate interventions span-	138
086	ning prompting, language-side adaptation, targeted	139
087	data construction, visual-side adaptation, representa-	140
088	tion probing, and explicit object-centric decomposition.	141
089	Structured prompting significantly hurts performance;	142
090	general and targeted LoRA improve aggregate accuracy	143
091	but do not remove the orientation ceiling; hard nega-	144
092	tives improve targeted relational coherence without a	145
093	significant overall orientation-accuracy gain; and the	146
094	tested visual-side LoRA configurations do not outper-	147
095	form the language-side control.	148
096	As a sensitivity analysis, removing examples	149
097	flagged by the single-annotator audit raises orienta-	150
098	tion accuracy (65.7%→78.5% for 7B General LoRA,	151
099	66.4%→76.6% for hard-negative LoRA on the strictest	152
100	107-example subset) but does not eliminate error (2B	153
101	structured shows no gain); a second blind annotator's	154
102	agreement with the original flags is assessed in the	155
103	supplementary. Because the removed examples were	156
104	drawn from persistent failures, part of this gain is ex-	157
105	pected by construction.	158
106	A second failure mode emerges from logical consis-	159
107	tency. The zero-shot 7B model is consistent on only	160
108	36.9% of complementary facing/facing-away pairs. LM-	161
109	side LoRA substantially improves coherence, and hard-	162
110	negative training raises facing consistency to 77.7%	163
111	while facing accuracy remains unchanged at 68.9%.	164
112	The extra hard-negative gain is not significant in the	165
113	pooled strict-family comparison, yet accuracy stays un-	166
114	changed while consistency moves: the two metrics cap-	167
115	ture different behavior.	168
116	Finally, we test cross-dataset transfer. The 7B	169
117	model is strong on SITE's official spatial-relationship-	170
118	reasoning category (75.1% raw; 59.2% chance-adjusted	171
119	accuracy), yet the VSR-trained adapter does not	172
120	generalize cleanly: it is statistically neutral overall	173
121	(54.2%→53.8%, $p = 0.66$) and significantly degrades	174
122	the official category (75.1%→71.6%, $p = 0.004$). A	175
123	pre-specified, non-official orientation-vocabulary slice	176
124	is much weaker in the aggregate (48.3% raw; 24.0%	177
125	chance-adjusted) and shows no positive transfer, but—	178
126	as we detail below—its composition limits what it can	179
127	establish, and we therefore do not treat SITE as inde-	
	pendent confirmation of the VSR orientation construct.	
	Our contributions are fourfold, each stated as a	
	reusable evaluation practice rather than as a model-	
	specific finding:	
	• A fine-grained diagnostic ladder. Across ten VSR	
	conditions and two differently sized models, sev-	
	eral spatial families improve substantially while	
	object-relative orientation remains unusually resis-	
	tant. The ladder—scaling, prompting, language-	
	side adaptation, targeted data construction, visual-	
	side adaptation, representation probing, and ex-	
	PLICIT decomposition—is defined operationally so it	
	transfers to other benchmarks and relation families.	
	• A robustness and sensitivity audit. A single-	
	annotator clean-label sensitivity analysis, frozen/object-grounded probes, visual-side adapta-	
	tion, and object-centric decomposition constrain	
	purely annotation-based or single-component ex-	
	planations. The audit practice (flagged-example	
	exclusion sets, decodability checks against majority	
	baselines) applies to any benchmark whose labels	
	or examples may be ambiguous.	
	• An accuracy-coherence separation protocol.	
	Complementary-relation tests on VSR's strict	
	relation pairs show that, for the tested conditions,	
	relational consistency can change substantially	
	while task accuracy remains fixed. The protocol	
	measures consistency against verdict-pattern base	
	rates, so it can be applied wherever complementary	
	formulations of the same relation exist.	
	• A cross-dataset transfer protocol. SITE shows	
	strong official spatial-relationship performance,	
	while VSR-trained LoRA transfers poorly and sig-	
	nificantly harms the official SITE category; the	
	orientation-vocabulary slice is analyzed as ex-	
	ploratory evidence. The protocol (pre-specified	
	subsets, chance-adjusted accuracy, paired transfer	
	tests) is model- and benchmark-agnostic and is re-	
	ported with negative as well as positive transfer.	
	None of these practices is specific to orientation or to	
	the two model families studied here; each is specified	
	operationally in the supplementary so that other re-	
	searchers can apply the same audit to other capabili-	
	ties, benchmarks, and models.	
	2. Related Work	
	Spatial reasoning benchmarks. VSR introduced a con-	
	trolled natural-image benchmark of more than 10k	
	image-text pairs spanning 66 spatial relations and ex-	
	PLICITLY highlighted orientation relations such as fac-	
	ing and parallel as difficult for the VLMs studied at	
	the time [7]. Our work should therefore be read as a	
	modern diagnostic extension rather than the first ob-	

180	ervation of an orientation weakness. What'sUp iso-	rank updates [5], to compare general, targeted, hard-	231
181	lates basic spatial relations by holding object identi-	negative, projector, and vision-plus-projector interven-	232
182	tities fixed while varying configurations, and found large	tions. This design lets us ask whether benchmark	233
183	gaps between humans and tested vision-language mod-	gains reflect general spatial competence or specializa-	234
184	els [6]. GSR-Bench extends controlled grounded spa-	tion. The external SITE transfer experiment is cen-	235
185	tial evaluation across a larger set of multimodal mod-	tral here: VSR adaptation is neutral overall on SITE,	236
186	els and model scales [9]. SITE broadens evaluation	does not improve the orientation subset, and signifi-	237
187	to single-image, multi-image, and video inputs and or-	cantly degrades SITE's official spatial-relationship cat-	238
188	ganizes tasks using cognitive-science dimensions includ-	egory. The result complements benchmark work by	239
189	ing spatial visualization/orientation, intrinsic/extrinsic	explicitly measuring negative as well as positive cross-	240
190	relations, and static/dynamic settings; its authors re-	dataset transfer.	241
191	port spatial orientation as a persistent challenge [14].		
192	Recent broad evaluations such as SpatialScore simi-	Representation probing and logical consistency. Di-	242
193	larly argue that spatial intelligence remains fragmented	agnostic probes can reveal whether information is ac-	243
194	across tasks and modalities [15], and Mind the Gap	cessible from frozen representations, but probe perfor-	244
195	isolates spatial reasoning from detection and semantic	mance must be interpreted cautiously because probe	245
196	comprehension in contemporary VLMs [10].	capacity can itself learn the target task [4]. We there-	246
		fore use linear and nonlinear probes, compare against	247
197	Diagnosing spatial failures. Recent work has begun	majority baselines, add object-grounded features, and	248
198	to analyze spatial failures mechanistically rather than	avoid claiming that weak decodability proves informa-	249
199	only benchmarking them. On VSR itself, AdaptVis	tion is absent. Separately, consistency across trans-	250
200	traces attention distributions and finds that success-	formed or complementary statements has prior litera-	251
201	ful spatial reasoning correlates with attention align-	ture, including consistency-oriented training objectives	252
202	ing to actual object locations [2]. CREG charac-	such as SAGE's geometric and linguistic duality oper-	253
203	terizes the directional organization of attribution evi-	ations [8]. Our consistency analysis is a VSR-specific	254
204	dence in Qwen2-VL on spatial-relation benchmarks,	diagnostic rather than a new consistency framework:	255
205	showing that standard heatmaps often reflect image	it uses strict complementary relation pairs with ob-	256
206	layout rather than the queried relation [11]. When	served verdict-pattern and base-rate analysis, and it	257
207	More Is Less analyzes when injecting spatial cues, com-	shows that instance-level accuracy and complementary-	258
208	monsense knowledge, or chain-of-thought instructions	relation coherence can diverge for the tested conditions.	259
209	helps or hurts VSR, with effects varying by model and		
210	setup [1]. Probing Visual Concepts measures linear	3. Experimental Setup	260
211	decodability of visual concepts, including orientation-		
212	relevant ones, in lightweight driving VLMs and identi-	3.1. Benchmarks	261
213	fies information-flow bottlenecks [12]. VSP decomposes		
214	visual spatial planning into perception and reasoning	VSR. We use VSR [7] as the primary diagnostic	262
215	subtasks [16], and MM-Spatial targets 3D spatial un-	benchmark. The random-split test set contains 2,195	263
216	derstanding with geometric inputs [3].	examples. The benchmark defines 64 relation labels;	264
217		61 appear in the evaluated test split. The orientation	265
218	Prior work has thus already established spatial and	family contains facing, facing away from, parallel to,	266
219	orientation weaknesses in VLMs and has begun to ana-	and perpendicular to, totaling 137 test examples. All	267
220	lyze those failures through prompting, attention, repre-	main VSR results use the same held-out test set and	268
221	sentation probing, and consistency-based training. Our	deterministic True/False parsing.	269
222	contribution is therefore not the first identification of		
223	orientation difficulty or spatial inconsistency. Instead,	SITE. We use SITE [14] for cross-dataset transfer	270
224	we revisit the known VSR orientation weakness in mod-	evaluation only and do not train on SITE. The image	271
225	ern generative VLMs and subject the same relation fam-	evaluation contains 2,591 examples: 1,368 single-image	272
226	ily to a controlled diagnostic ladder spanning adapta-	and 1,223 multi-image examples. Our primary SITE	273
227	tion, frozen and object-grounded readouts, visual-side	subset is the official spatial-relationship-reasoning cate-	274
228	interventions, complementary-relation consistency, la-	gory ($n = 993$). We additionally report a transparently	275
	bel sensitivity, and cross-dataset transfer.	defined, non-official keyword-derived orientation subset	276
229	Adaptation and spatial specialization. We use LoRA,	($n = 2,024$), analyzed as exploratory evidence rather	277
230	which adapts frozen models through trainable low-	than as direct replication of the VSR construct. SITE	278

results use the benchmark’s native multiple-choice format and chance-adjusted accuracy (CAA) in addition to raw accuracy.

3.2. Models and Conditions

Our compact model is SmolVLM2-2.2B-Instruct, and our larger model is Qwen2-VL-7B-Instruct [13]. We evaluate ten canonical VSR conditions: 2B zero-shot, 2B structured prompting, 2B General LoRA, 2B Targeted LoRA, 7B zero-shot, 7B General LoRA, 7B Targeted LoRA, 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B Vision+Projector LoRA. LoRA [5] is used for all parameter-efficient adaptation conditions.

General LoRA uses a proportional sample across relation families. Targeted LoRA allocates most training examples to weak orientation/depth/horizontal families with rehearsal from the remaining relations. Hard-negative LoRA introduces audited complementary orientation examples, primarily facing/facing-away inversions. Vision-side conditions adapt either the multi-modal projector or upper vision blocks together with the projector. The frozen experimental snapshot records the final result set.

3.3. Metrics and Statistical Tests

For VSR we report accuracy overall and by spatial family. For SITE we report raw accuracy and CAA. Paired condition comparisons use McNemar tests when applicable. Clean-label orientation analysis reports the full orientation set and progressively stricter audited subsets. Logical consistency is evaluated by constructing complementary statements for strict relation pairs (left/right, front/behind, facing/facing-away) and testing whether the model assigns opposite verdicts as required. Parallel/perpendicular is treated separately as a soft complement because both relations can be false for oblique objects.

4. Main Results

4.1. Orientation Remains Weak Across Two Differently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B model to the 7B model lifts zero-shot overall accuracy by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and containment by 5.9 pp, but orientation by only 0.7 pp. General 7B LoRA further raises overall accuracy to 84.7%, yet orientation reaches only 65.7%. Across trained conditions other than structured prompting, orientation remains in a narrow 62–66% band.

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while orientation falls from 62.8% to 53.3%. A paired exact

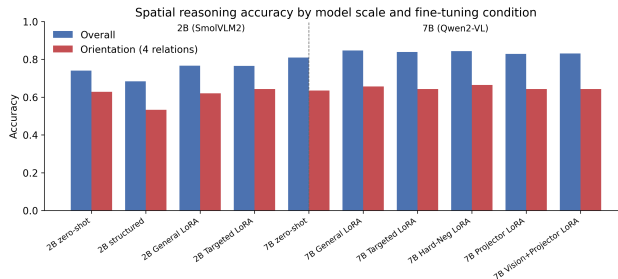


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Consistent with recent analyses of information injection on VSR, where additional spatial cues or reasoning instructions can help or hurt depending on the model and setup [1], this result is specific to the tested 2B condition and prompt; it does not establish that prompting generally fails.

4.2. Clean-Label Sensitivity Analysis

A single-annotator exploratory audit identified ambiguous or questionable orientation examples, so we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these values should not be read as a matched comparison against other relation families. A second, independent blind annotator re-rated the same examples: agreement with the original flags is only fair ($\kappa=0.36$, 66.7% agreement, on the clean/ambiguous flag; Krippendorff’s $\alpha=0.14$ on the eight-class taxonomy), so the exclusion subsets are annotator-specific; the frozen table is retained rather than redefined, and an annotator-dependence analysis is in the supplementary. The robust conclusion is narrower: removing flagged examples raises orientation scores but does not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations without fine-tuning the generative model. Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4

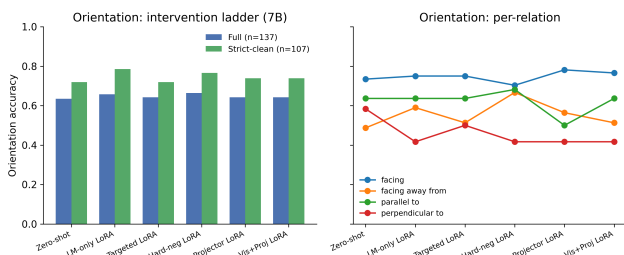


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ($n=34$) and the confidence interval overlaps the majority baseline. In short: orientation is not robustly decodable by the tested frozen-feature readouts.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Linear decodability of visual concepts has been probed in driving-oriented VLMs [12], and directional information in this model family has been inspected via attribution [11]; our probe analysis is not the first of either kind. It differs by targeting VSR’s orientation relation family with pooled, patch-

vote, and object-grounded readouts over the exact conditioning features of the generative model, and by pairing decodability with adaptation and consistency interventions rather than treating it as an end in itself. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted $p=0.0043/0.0122$; these effects do not survive a blanket multiple-comparison adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not unlock the ceiling.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s True/False verdicts agree with the end-to-end control on only 55.1–58.3% of the 127 boxed orientation statements; in the committed analysis (10 no-box statements scored wrong), verdicts diverge on 61–65 statements, and the control is correct while the pipeline is not in 44–47 of them. The committed pipeline scored agreement with the control rather than ground-truth accuracy, so we report the tested decomposition as failing to reproduce the generative model’s behavior, not as a direct accuracy comparison. Two-

dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair and transformation-based consistency evaluation has prior literature, including consistency-oriented training [8]; our contribution is the VSR-specific diagnostic below, built on strict complementary relation pairs with observed verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The observed verdict patterns show what drives this: the model affirms either member of a facing pair only 17–20% of the time, a False-defaulting bias that yields both-False verdicts on 63.1% of pairs and no both-True verdicts (0/103); expected consistency under verdict independence given these base rates is 30.2%, below the observed 36.9% (verdict table in the supplementary): the dominant phenomenon is a refusal to affirm either proposition, rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind consistency to 70.7%; pooled strict-family consistency is significantly better than zero-shot ($p < 0.0001$). Hard-negative training further raises facing consistency to 77.7%. Facing accuracy, however, is identical for LM-only and hard-negative LoRA at 68.9%. The additional hard-negative improvement is not significant in the pooled strict-family McNemar comparison ($p = 0.29$). Paired hard-negative supervision therefore provides a targeted intervention on complementary-relation coherence: it descriptively raises facing-pair consistency while leaving original facing accuracy unchanged, although the additional pooled consistency effect over LM-only is not significant. We do not claim a general coherence algorithm; the key result is separability: identical facing accuracy coexists with substantially different facing consistency. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

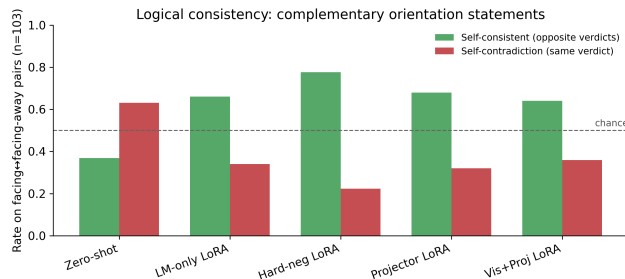


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-choice protocol [14], with no SITE training. Because the pre-specified decision rule resolved to branch 2 (the official category was not broadly weak), this adapter run is a post-protocol cross-dataset transfer check rather than a pre-specified step; the protocol and its outcome are documented in the supplementary. The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer on the independent benchmark; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified, non-official orientation-vocabulary slice is substantially weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category, and below the 54.2% / 31.1% all-images average) and shows no positive transfer under the adapter (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship category), we treat it as exploratory evidence rather than as direct replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI [0.60, 1.16], $p = 0.29$), although it remains descriptively 7.5pp lower within the official

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	-0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	-3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4



Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

spatial-relationship category itself (71.3% vs. 78.8%, $n=488/505$). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; $n=177$) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation that orientation relations are difficult [7], while extending it to modern generative VLMs and a much broader intervention analysis.

Decodability and coherence both matter, as diagnostics. The probes show that orientation is not robustly decodable under the tested frozen-feature readouts, while consistency experiments expose large logical failures even for relations answered above chance. Neither result alone identifies a complete causal mechanism,

and both are diagnostic rather than causal-localization claims. Together they support a mixed account in which orientation information is difficult to access reliably under the tested readouts and relational decisions are not always propagated coherently across complementary formulations.

Task adaptation can specialize rather than generalize. General LoRA improves VSR overall accuracy substantially, but its SITE transfer is neutral overall and significantly negative on the official spatial-reasoning subset. Hard negatives improve a targeted coherence behavior while producing relation-specific tradeoffs. Benchmark-specific fine-tuning gains should therefore not be equated automatically with general spatial-reasoning improvement.

8. Limitations

Statistical power is limited at the scale of the orientation diagnostic. The VSR orientation family contains only 137 test examples, and the smallest per-relation subset, perpendicular, has $n=12$; perpendicular results are therefore reported descriptively throughout. The Wilson 95% confidence intervals on the family-level point estimates are correspondingly wide—62.8% [54, 70], 63.5% [55, 71], 65.7% [57, 73], and 66.4% [58, 74] for 2B zero-shot, 7B zero-shot, 7B General LoRA, and 7B Hard-Neg LoRA ($n=137$ each)—so the trained conditions’ 62–66% band should be read as a set of statistically overlapping point estimates rather than a fine-grained ranking. Several smaller point estimates have confidence intervals that include the majority baseline or chance: the patch-vote parallel/perpendicular probe (73.5%, $n=34$; 95% CI [57, 85] vs. the 64.7% majority), the perpendicular subset for the trained 7B conditions (41.7%, $n=12$; [19, 68] vs. 50%), and the 7B zero-shot facing-away subset (48.7%, $n=39$; [34, 64]). Paired McNemar tests on the orientation family rest on small discordant-pair counts and have limited power: the non-significant projector and vision+projector orientation comparisons ($p=0.86$, $p=0.85$) indicate an absence of detectable effect, not evidence of equivalence. Where the paper draws conclusions, it relies on the full-set paired tests and the Holm-adjusted battery in the supplementary rather than on these small- n point estimates alone.

The clean-label audit rests on one annotator’s flags: a second blind annotator reached only fair agreement on the clean/ambiguous flag ($\kappa=0.36$; Krippendorff’s $\alpha=0.14$ on the eight-class taxonomy), so the exclusion subsets are annotator-specific, and the sensitivity table is reported as frozen rather than redefined. The SITE

orientation slice is keyword-derived rather than an official benchmark category; a composition-confound analysis shows its aggregate score is largely explained by task composition, so we do not treat SITE as independent confirmation of the VSR orientation bottleneck. Our probing methods test accessible decodability under specific linear/MLP and pooling choices; failure to decode does not prove information is absent. Visual-side experiments use LoRA rather than full end-to-end retraining. We study two differently sized VLMs from different families; this is a cross-model comparison, not a controlled scale experiment. All trained conditions are single-checkpoint runs: the paired McNemar tests quantify disagreement on the fixed test examples, not training-run variability, and these single-seed results do not estimate it. The SITE evaluation is image-only; SITE video reasoning remains future work. Finally, several interventions are evaluated as negative results, so lack of improvement should be interpreted for the tested configurations rather than as impossibility claims about all prompting, adaptation, or architectural methods.

9. Conclusion

Aggregate spatial accuracy hides a persistent orientation bottleneck in the VLMs studied here. A larger VLM and standard adaptation substantially improve several spatial families while leaving object-relative orientation comparatively resistant. Auditing, probing, visual-side adaptation, and decomposition constrain simple explanations, while complementary-relation tests show that accuracy and coherence are distinct axes of spatial competence. Cross-dataset evaluation on SITE further shows that VSR-specific adaptation transfers poorly and can negatively transfer to official spatial-relationship reasoning. The results motivate spatial evaluation that reports relation-specific performance, transfer, and logical coherence rather than relying on aggregate accuracy alone.

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